Decision Analytics Journal 9 (2023) 100346

Contents lists available at ScienceDirect

Decision Analytics Journal

journal homepage: www.elsevier.com/locate/dajour

An analysis of expert-opined strategies in multi-attribute decision-making using effort propagation

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ARTICLE INFO

Keywords:
Multi-attribute decision-making
Inter-relationships
Hierarchical structures
Propagation strategy
Effort propagation analysis
Performance assessment

ABSTRACT

Mining the interrelationships and possible hierarchical structures among the factors in real-world multi-attribute decision-making (MADM) problems is considered one of the primary tasks. One major task is determining an optimal strategy to work on the factors to enhance the effect on the goal attribute. This paper proposes two such strategies: parallel and hierarchical effort assignment and propagation strategies. We formally define and describe the concept of effort propagation through a strategy. Both the parallel and hierarchical strategies are divided into sub-strategies based on whether the assignment of efforts to the factors is uniform or depends upon some appropriate heuristics related to the factors in the system. The adapted and discussed heuristics are the factors' relative significance and effort propagability. The strategies are analyzed for a real-life case study regarding high school administrative factors that enhance students' performance. Total effort propagation of around 7%–15% to the goal is seen across the proposed strategies, given a total of 1 unit of effort to the directly accessible factors of the system. A comparative analysis is adapted to determine the optimal strategy among the proposed ones to enhance student performance effectively. The highest effort propagation achieved in the work is approximately 14.4348%. The analysis establishes the necessity of research towards the direction of effort propagation analysis in case of decision-making problems.

1. Introduction

In many real-world fields such as business, health sector, educational institutes, etc., decision-making is one of the important tasks to handle. For instance, what are the important factors for better growth of a business, and what should be a proper strategy to enhance the growth? or what are the significant factors that play an important role in improving the performance of a student? Generally, we refer to the problem of handling these issues as Decision-Making problems. So, structurally, a decision-making problem consists of one or several factors related to a particular goal on which a decision is to be made. In most of the complex problems, we see the number of factors is more than one, in fact in some cases there may be a large number of factors contributing to the decision-making process. Such systems are generally called multi-attribute decision-making (MADM) systems. The primary tasks in the case of a MADM system are mining the interrelations among the deciding factors and determining their system significance, or discovering if there is any hidden hierarchy among the factors or not in terms of their significance. These tasks are mainly performed under the supervision of a group of experts chosen based on their subject expertise. For example, in a business decision-making process, the experts can be the director, manager, and several other higher-rank officers, or for a particular project work, they can be the project supervisor along with several project heads. In health sectors, a particular decision-making process can be supervised by a group of expert doctors, or in the education sector, a group of teachers, professors, or other administrative higher authority can act as the experts. Hence, such decision-making problems can be thought of as an expert-opinion mining task also related to the particular decision to be made. The concerned decision-making methods are called multi-attribute group decision-making (MAGDM) methods [1–5].

In literature, the most popular MAGDM method to handle the task of mining the inter-relationships among the deciding factors in the system is the 'decision-making trial and evaluation laboratory' (DEMATEL). The method was first conceptualized by Gabus and Fontela [6,7]. Then, several of its other upgraded versions such as Fuzzy DEMATEL [8,9], Intuitionistic Fuzzy-DEMATEL [10,11], etc. did the same task but with more precision. To incorporate the vagueness of the expert opinions, these methods made use of fuzzy [12] or intuitionistic fuzzy [13] linguistics. The methods captured the causal relationships among the factors, i.e., after analyzing the hidden inter-relations among the factors, the factors were classified into two groups "cause" and "effect". In 2023, Dhara and Goswami [14] discussed some structural issues of these methods and came up with a novel decision-making method, called 'causal relationship and ranking technique' (CRRT) which not

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https://doi.org/10.1016/j.dajour.2023.100346
Received 11 August 2023; Received in revised form 20 September 2023; Accepted 22 October 2023
Available online 27 October 2023
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only mined the inter-relationship among the factors but by finding their system significance, the method ranked the factors as well. To discover a possible hierarchy among the factors, methods like 'interpretive structural modeling' (ISM) [15,16], 'analytic hierarchy process' (AHP) [17,18], 'analytic network process' (ANP) [19,20], etc. were handy. However, Dhara and Goswami showed that in dense systems, i.e., where the factors are densely interrelated, the method ISM completely failed to establish a proper hierarchy among the factors. They also came up with the concept of an iterative hierarchy method, called 'iterative hierarchy and ranking process' (IHRP) [21]. The method iteratively finds out the hierarchical structure among the factors, even if they are densely interconnected. We have discussed the working procedure and limitations of the methods in more detail in the "Literature Review" section.

• **Research Objective and Motivation**

We see that in decision-making systems, although the determination of hidden inter-relationships or a probable hierarchy among the factors helps immensely in the decision-making process, there remains a vital question to be answered which is 'given the inter-relations or the hierarchy what should be the optimal strategy to act upon the factors to obtain the best results in terms of decision-making?'. Sometimes, the number of possible strategies for reaching the best state of the goal is finite. For example, in deciding the value-for-money item or a product based on some of its features or factors, the number of criteria to analyze for the factors (that are strategies in such cases) is limited. Various standard out-ranking techniques such as the 'technique for order preference by similarity to ideal solution' (TOPSIS) [22,23], or 'visekriterijumska optimizacija I kompromisno resenje' (VIKOR) [24,25] do the best to solve such tasks and finds out the best strategy for the goal optimization. But, in many decision-making systems, the goal is abstract and there can be too many or sometimes infinitely many strategies for goal optimization. For instance, consider the goal of enhancing the health facility in the health sector or improving student performance in the education sector. In these cases, the goal can be optimized based on the efforts assigned to the various related factors, and this assignment of efforts to the factors can be done in various ways. In such situations, 'how much effort is put into the factors and in which pattern' is a critical analysis to make to get the best results for the concerned decision-making process. Although there are many methods and analyzes in the literature that handle the task of relationship mining and hierarchy determination, to the best of our knowledge, the existing research work from the perspective of strategic analysis for a decision-making problem is void. Hence, the main objective of this paper is to analyze various strategies in terms of effort assignment to the factors so that the main goal of the decision-making system is maximally enhanced. We have formally defined the research problem in detail in the "Problem Definition" section. In this paper, we propose and analyze two such strategies to enhance the effect on the goal attribute based on a novel concept of effort propagation. Since we have tried to analyze various strategies for the enhancement of the objective of the decision-making problem from the perspective of effort assignment to the factors and analyzed the effort induction/propagation in the system, we term the analysis as "Effort Assignment and Propagation Analysis". Moreover, the strategies were discussed and analyzed for a real-life case study regarding the Indian high school administrative factors that play an important role in the enhancement of students' performance. In this regard, it is a must mention that in the field of educational data mining (EDM), lots of prior research work for the problem of enhancing student performance has been done from the perspective of student data, which includes their academic data, demographic, behavioral, non-behavioral data, etc. But we think that to enhance student performance, along with the student data, several factors of the administration play an important role too and these relationships can only be mined with the help of expert opinions. Unfortunately, studies from the perspective of experts in this field are very limited and need more attention. Hence, this work, by adapting the case study of enhancement of student performance based on the expert-opined administration data tries to enrich the analysis of student performance enhancement from the expert/administration point-of-view too.

The main contribution of the paper is (1) we propose and analyze two different types of effort assignment and propagation strategies, namely parallel and hierarchical strategies, to enhance the goal of a decision-making problem maximally, (2) the undertaken case study of analyzing the strategies in the case of Indian high school administrative factors affecting students' performance can be thought of as a novel approach of expert-opinion mining in the field of student performance assessment/prediction too. The remainder of the paper is arranged as follows. In Section 2, we have surveyed the existing literature for various related MAGDM methods and discussed the necessity of the strategy analysis in more detail. In Section 3, we formally define the problem of effort assignment and propagation. In Section 4, the detail of the proposed strategies and their functionality is discussed. Section 5 presents the description of the undertaken case study. The paper concludes with Section 6 which discusses the future study aspects of this work as well.

2. Literature review

This paper intends to explore any real-life decision-making problem from a strategic point of view. That is, it tries to analyze various strategies for assigning efforts to the factors of the system so that the propagated effect to the goal attribute for that particular effort assignment is maximized. The effort assignment to the factors depends upon various criteria within the system. For example, the inter-relationship among the considered factors can be an important aspect to look up to before going on for the effort assignment. In the existing literature, there are various MAGDM methods for mining the inter-relationship among the factors in terms of the inter-factor influences, but the most popular among them is the DEMATEL method, originally invented by Fontella and Gabus [6,7]. Since then, the method [26–28] has been in extensive use for several real-life decision-making problems. The method starts with the opinion matrices on the inter-factor influences evaluated by several experts in the field under consideration. Then, by aggregating all the opinion matrices, it determines the total influence among the factors of the system and checks which influence values are significant and which are not significant on the basis of some threshold value. The threshold value can be pre-fixed or can be obtained from the data. Finally, based on the influence and dependence values of the factors on the system, the factors are classified into the "Cause" and "Effect" groups. Several extensions [29–31] of the DEMATEL framework advance the working procedure of the conventional DEMATEL method by determining the system significance values of the factors based on their system prominence and relation values [32]. We see that the expert evaluations for the inter-factor influences depend solely on the subject expertise of the decision-makers in that field and there may lie some amount of uncertainty or vagueness in those evaluations. To cope with this situation, the use of fuzzy linguistics for the DEMATEL approach came into the picture. Several methods such as Fuzzy-DEMATEL [33,34], Intuitionistic Fuzzy-DEMATEL [35,36], etc. performed the same task of mining the inter-factor relationships but using the fuzzy, intuitionistic fuzzy linguistics for the inter-factor influence evaluations increased the accuracy of the methods. In 2023, Dhara and Goswami [14] detected out some structural limitations of the conventional and fuzzy-integrated DEMATEL methods. One of the major drawbacks of the existing methods was the convention used to classify the factors as "cause" and "effect". The existing methods classify the factors based on the degree of influence and dependency of the factors on the system. They pointed out that there may be some situations, where a factor may influence a smaller number of factors than any other factor, but the amount of total influence of the former factor may be high compared to the latter one. In this case, there should

be some tradeoff option for the decision-maker to decide whether the former factor belongs to the "cause" group or the "effect" group, and the same for the later factor as well. Keeping this in mind, they came up with a novel CRRT framework [14], which introduced the use of a priority parameter 'p' to trade off the decisions of the experts in the aforementioned situations. The method also introduces two separate causal relationship diagrams, namely the influence and dependence network, and self-group network depicting the inter-factor relationships within the system. The method also captures the system significance values for all the factors and ranks them.

Another important aspect before proceeding with the strategic analysis is the hidden hierarchy among the factors of the system. A hierarchy among the system presents a useful overview of the system at first glance and the decision-makers can tentatively sense which factors to emphasize while putting on the efforts and which are the factors that can be neglected or given less importance. The concerned methods in the literature are AHP [37,38], ANP [39,40], ISM [41,42], etc. All these methods work with the pair-wise comparison matrix between the factors and try to determine the underlying hierarchy among the factors. As in the case of inter-factor relationship mining, here also, the use of fuzzy linguistics can be found in the literature [43–45] in order to mine the vagueness in pairwise comparison of the factors. One of the main problems with these methods was the inability to build a proper hierarchy among the factors when they are highly interrelated. We call such systems as "dense" systems. In 2023, Dhara and Goswami addressed this issue with the ISM framework and proposed an iterative hierarchy-building framework, named IHRP [21]. The method is divided into three stages in which the first stage is dedicated to mining the inter-relations among the factors from the expert opinions whereas the second and third stage is there to determine a proper crisp hierarchy among the factors and find their ranking by determining their significance in the system. The basic intuition for the hierarchy building in this framework is to place factors that have lower dependency on the lower sub-level factors and higher influence on the upper sub-level factors in relatively lower sub-levels of the hierarchy. Factors that have the same number of influent factors were placed in the same level and depending upon the number of their influenced factors in the higher levels, further sub-leveling is done iteratively. A typical sub-level for a factor in the IHRP-produced hierarchy looks like "II-C", which means that the factor is in the third sub-level (C) of the second level (II). Also, upon building a proper crisp hierarchy among the factors, in the third stage, the system significance values of the factors are determined based on their position in the hierarchy and their drive (influence) and dependence values. The method is able to work effectively in the case of dense systems.

Once the relationships among the factors and the possible underlying hierarchy among the factors are determined, we propose that a proper strategic investigation is highly needed in order to maximally enhance the objective of the decision-making problem. Unfortunately, to the best of our knowledge, there is no systematic literature for decision-making problems focusing on the strategic analysis aspect. Hence, this accounts for the novelty of the approach taken in this paper. In the next section, we formally define the problem under consideration in this paper.

3. Problem definition

This paper deals with the problem of appropriate effort assignment to the factors of a system so that the goal attribute gets maximally affected. We can define "**Effort**" in terms of the allotment of funds, manpower, or assignment of necessary infrastructures, logistics, etc. Measuring the enhancement of the goal attribute is not always feasible instantly and can be a time-consuming process. For theoretical studies, the enhancement of the goal sometimes needs real-time validation. But, in any case, we can think of 'the amount of effort that gets propagated to the goal attribute' to be a good estimator of the measurement of the goal enhancement. The basic intuition is that the more effort gets propagated to the goal, the more it is affected/enhanced. Hence, in our study, we use an 'index of effort propagated to the goal' as the measure for the enhancement of the goal attribute in the decision-making system. The definition and calculation of the index are described in the later section. In a system, where the factors are interconnected through hidden relationships/ inter-influences, the allotted effort to a particular factor propagates through various existing paths to other factors, and ultimately some portion of this effort reaches and affects the goal. In decision-making problems, there can be a number of possible assignments of effort to the factors, and they have different impacts on the goal attribute. But in a decision-making environment, our main objective is to enhance the goal maximally. Hence, it becomes quite necessary to determine a proper strategy for this effort assigning, i.e., working on factors so that the goal gets maximally enhanced. For this, we need to find proper strategies for assigning efforts to the factors of the system, so that, the amount of effort propagated to the goal attribute is maximum, which in turn can result in the optimal enhancement of the objective/goal. For instance, one criterion of assigning a certain amount of effort to a particular factor in the system can be the attribute weight or the system significance value of the factor. There may be a little bit of confusion between "effort" and "attribute weight" since they are both associated with the factors. But one needs to understand that "efforts" are nothing but some monitory, or logistical assignment to the factor based on its attribute weight as one criterion and the attribute weight for a particular factor is the relevance of the factor which is determined in terms of its influence and dependence values in the system by several aforementioned MAGDM methods. There may be many other criteria for the effort assignment. A pre-analysis of the strategies rather than randomly working on the factors sometimes saves a lot of time and also informs us of the best-estimated strategy for achieving the best state of the main objective/goal. In this paper, we propose two expert-opinion-led strategies of assigning efforts to the factors and analyze the effect on the goal attribute using the concept of Effort Propagation and try to find the optimal strategy among them. The formal definition of the problem under consideration is given below.

• ***Problem of finding an optimal strategy to maximize the enhancement of the Goal attribute of the Decision-Making problem***

Given factors $F_1; F_2; \ldots\ldots\ldots\ldots F_n$ in a decision-making system, the problem is to find out the optimal strategy of assignment of efforts to the factors, that corresponds to the maximum effort propagation to the Goal.

In the next section, we describe two strategies of effort assignment and a novel concept of Effort Propagation mechanism in detail.

4. Methodology

For a system where the factors are interconnected through significant influences, there may lie various strategies for effort assignment to the factors which in turn affects the goal in different ways or proportions. In this work, we describe two such strategies and compare these strategies in terms of their effort propagation capabilities. But before analyzing the effort propagation in such strategies, we need to determine the inter-relationships among the factors present in the system and determine the relative significance of the factors in the system. To do that, we can take the help of expert opinions on the factor inter-influences and deploy any of the existing MAGDM techniques [46]. We briefly describe the process of mining inter-relationships among the factors in the next subsection.

4.1. Mining inter-relationship and system significance of the factors

In a system consisting of factors that are interlinked through inter-influences, we can take the help of expert opinions on the factor inter-influences. For that, we need to conduct expert interviews with a questionnaire or approach them with the list of factors and ask them

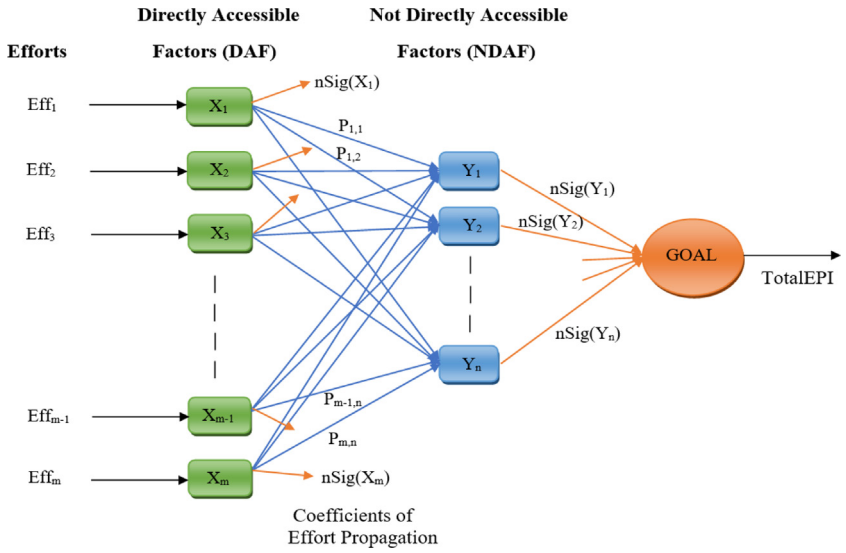

Fig. 1. Schematic diagram of the PEAP strategy.

to share opinions on the influence of one factor on another based on some prefixed linguistic scale of influence. If there are n factors in the system, then the opinion data from each expert is in the form of a $n \times n$ matrix. Next, we can deploy any of the MAGDM methods that exist in the literature. The applicable methods for mining the inter-influences of the factors within the system and their system significance values are DEMATEL [47,48], CRRT [14], ISM [15], IHRP [21], etc. All these methods calculate the weighted aggregation of all the opinion matrices and determine a total relation matrix (TRM). The entry $t(i \to j)$ of this matrix represents the total influence the factor-i has on factor-j. Based on the entries of the TRM, (average of the entries, or sum of the mean and standard deviation of the entries, etc.) a particular threshold value is obtained and the values exceeding the threshold are considered as significant, and other values are considered as insignificant. Schemes like DEMATEL, and CRRT dig deeper from this matrix to find several aspects of the inter-factor influences to determine the system significance values of the factors. On the other hand, ISM, IHRP, etc. are hierarchical frameworks that determine the underlying hierarchy among the factors and then determine their system significance values based on their hierarchical positions and inter-influence structures. In the case of dense systems, i.e., systems where the factors are densely inter-influenced, the ISM method fails to determine a proper hierarchy [49], but the IHRP method performs well. In the next subsection, we propose and describe two effort assignment strategies and discuss the effort propagation in those strategies. The inter-influence and significance values of the factors will be actively used to find the total effort propagation in those strategies.

4.2. Proposed strategies

We propose and analyze two different possible strategies as below.

1. Parallel Effort Assignment & Propagation (PEAP) Strategy
2. Hierarchical Effort Assignment & Propagation (HEAP) Strategy

4.2.1. Parallel effort assignment & propagation (peap) strategy

Among the factors present in a system, some factors are not directly accessible, i.e., we cannot put effort into the factors directly. We can call such factors Latent Factors or Not Directly Accessible Factors (NDAF). Other factors can be directly worked on. Such factors can be referred to as Directly accessible factors (DAF). In the Parallel strategy, we assign efforts to all the directly accessible factors simultaneously and analyze the direct effort propagation to the goal attribute as well as indirect effort propagation to the goal through the latent factors. We assume that effort propagation from one factor to another requires a minimum time gap. Hence, no effort induction/propagation happens between the directly accessible factors as they are worked on simultaneously. Also, we assume that the effort propagation works only when all the significant directly accessible factors are worked on together, i.e., the whole system of factors is put into effort. Fig. 1 depicts the working mechanism of the PEAP strategy.

The steps for calculation of the total effort propagation in this strategy can be listed as follows

STEP 1. Assign efforts to the directly accessible factors. Let, the assigned effort to the factor F_i be Eff_i.

STEP 2. If F_j is a latent factor, then the propagated effort from F_i to F_j is calculated by the formula given below.

$$EP_{i \to j} = Eff_i \times d(i \to j) \tag{1}$$

where $d(i \to j)$ is the normalized direct influence of F_i on F_j.

STEP 3. The effort propagation from the factors to the Goal is directly proportional to their normalized system significance values. Hence, the effort propagated to the goal from the factor F_i can be calculated as below.

$$EP_{i \to goal} = Eff_i \times nSig(F_i) + \sum_{F_j \in NDAF} EP_{i \to j} \times nSig(F_j) \tag{2}$$

STEP 4. The total effort propagation index in this strategy is thus calculated as follows.

$$TotalEPI = \sum_{F_i \in DAF} EP_{i \to goal} \tag{3}$$

The normalized direct influence values in Eq. (1) are obtained from the direct influence values between any pair of factors F_i and F_j as follows.

$$d(i \to j) = \frac{D(i \to j)}{\sum_j D(i \to j)} \tag{4}$$

where $D(i \to j)$ is the direct influence value of the factor F_i on F_j obtained after the fuzzy processing of the expert opinion matrices.

As a summary, we can calculate the total effort propagation index ($TotalEPI$) by the following matrix equation

$$TotalEPI = E_{DAF}^{T} \left[nSig_{DAF} + P_{DAF \to NDAF} \; nSig_{NDAF} \right] \tag{5}$$